

FULL STACK + AI/ML PROJECT

# SAVORA

SustainableFoodRescue&Ordering Platform

A technical proposal for a digital ecosystem connecting surplus food providers with budget - conscious consumers through intelligent demand prediction and dynamic optimization.

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# 1. Introduction

SAVORA is a full-stack web application designed to bridge the gap between food businesses with surplus inventory and consumers seeking affordable meals. By digitizing the "food rescue" process, the platform converts potential waste into revenue for businesses and value for customers.

*"Instead of allowing suitable surplus food to become waste, SAVORA enables businesses to list it at a discounted price with full transparency on preparation and safety."*

## 2. Problem Statement

Food wastage remains a critical global issue. Restaurants and hotels often face a disconnect between preparation and actual demand. Key challenges include:

- Loss of potential revenue from unsold, high-quality inventory.
- Lack of a dedicated platform for consumers to discover discounted food.
- Limited transparency regarding the remaining "safe-to-consume" window for surplus items.

## 3. Proposed Solution

SAVORA implements a **List → Understand → Predict → Recommend → Rescue** workflow. It provides an AI-enabled interface for businesses to upload surplus items with metadata including quantity, original/discounted price, best-before time, and allergen info.

## 4. System Learning & Data Analytics

The system evolves through permitted platform interactions to optimize the food rescue cycle:

| Category             | Data Points Tracked                                                                             |
|----------------------|-------------------------------------------------------------------------------------------------|
| Customer Preferences | Veg/Non-veg leanings, price sensitivity, preferred pickup times, and geographic range.          |
| Demand Patterns      | High-velocity items (e.g., Biryani vs. Salads), peak ordering periods, and discount elasticity. |

## 5. AI/ML Capabilities

The core intelligence of SAVORA is driven by several key machine learning models:

| Cap ability                  | Functional Purpose                                                                |
|------------------------------|-----------------------------------------------------------------------------------|
| Demand Prediction            | Predicts expected surplus levels based on historical trend analysis.              |
| Personalized Recommendations | Matches food listings to users based on budget, distance, and taste.              |
| Dynamic Discounting          | Suggests optimal price points based on remaining shelf life and quantity.         |
| Expiry Prioritization        | Algorithmic sorting to ensure food with the shortest safety window is seen first. |

## 6. The "Non-Trivial" Insight Requirement

A distinguishing feature of SAVORA is its ability to derive patterns beyond manual logic. The AI engine is tasked with identifying non-obvious correlations, such as:

- *"Surplus meals listed between 7:00 PM and 8:00 PM with 30–40% discounts have a 2x higher conversion rate than deeper discounts listed after 9:00 PM."*
- *"Customers within a 3km radius prioritize 'Time to Pickup' over 'Price' when the safety window is <2 hours."*

## 7. Technical Architecture

### Frontend

React.js / HTML5 / CSS3 / JavaScript for a responsive, interactive user experience.

## **Backend & API**

Python-based framework (Django/FastAPI) to handle business logic and RESTful communication.

## **AI/ML Stack**

Python (Scikit-learn, Pandas, NumPy) for model training and real-time inference engines.

## **Database**

Hybrid approach using MySQL/PostgreSQL for relational transactional data and MongoDB for flexible listing metadata.

## 8. Key Features & Workflows

### For Customers

- Map-based discovery of surplus food.
- Smart filters (Price, Distance, Dietary Needs).
- Real-time order tracking and secure payments.

### For Businesses

- Inventory dashboard with AI-suggested discounts.
- Sales analytics showing "Waste vs. Rescue" metrics.
- Automated listing expiration management.

## 9. Safety & Compliance

To ensure user trust, the platform enforces strict data fields for every listing: Preparation timestamps, storage requirements, and clear "Best Before" indicators. The system proactively delists items once they reach their safety threshold.

### SAVORA Project Proposal

Created for Sustainable Development & Technical Innovation

*"Don't waste food that someone can still value."*